Week 4 · Class 1: Practice Exercises — ANSWER KEY

Transforming Variables & Causality

2026-01-01

1 Segment A — Transforming and Creating Variables

2 Non-AI Exercises

2.1 1. Understanding mutate()

2.1.1 1.1 Multiple Choice: mutate() Function

What does the `mutate()` function do?

a) Removes columns from a data frame
b) Creates new columns or modifies existing ones
c) Filters rows based on conditions
d) Sorts data by a variable

Answer: **b) Creates new columns or modifies existing ones**

2.1.2 1.2 Code Detective: Basic mutate()

What does this code create?

```
data %>%
  mutate(
    vote_margin = dem_votes - rep_votes,
    winner = if_else(vote_margin > 0, "Democrat", "Republican")
  )
```

Line 3 creates: **A numeric variable showing the difference between Democratic and Republican votes (vote margin)** Line 4 creates: **A character variable indicating which party won based on the vote margin**

2.1.3 1.3 Fill in the Blanks: Variable Types

When creating new variables, we often need to:

1. Calculate **differences** between existing variables
2. **Recode** text variables into categories
3. Create **logical** (TRUE/FALSE) indicators
4. Convert between **data** types
5. Handle **missing** values appropriately

2.2 2. Conditional Logic

2.2.1 2.1 Match: if_else() vs case_when()

Match each function with when to use it:

Functions: a) if_else() b) case_when()

Use cases: 1. Creating a variable with only two possible outcomes 2. Creating a variable with multiple categories 3. Simple yes/no binary coding 4. Complex multi-condition logic

Matches: a = **1 and 3**, b = **2 and 4**

2.2.2 2.2 Code Detective: case_when()

What categories does this code create?

```
data %>%
  mutate(
    age_group = case_when(
      age < 30 ~ "Young",
      age < 50 ~ "Middle",
      age < 65 ~ "Older",
      TRUE ~ "Senior"
    )
  )
```

For someone age 25: **Young** For someone age 45: **Middle** For someone age 70: **Senior**

2.2.3 2.3 Spot the Error

What's wrong with this case_when() statement?

```
mutate(
  income_cat = case_when(
    income < 30000 ~ "Low",
    income < 50000 ~ "Medium",
    income < 30000 ~ "Low",
    TRUE ~ "High"
  )
)
```

Problem: **Line 5 duplicates the condition from line 3 (income < 30000). This is redundant and the second instance will never be reached because case_when() evaluates conditions in order.**

2.3 3. Working with Proportions

2.3.1 3.1 Multiple Choice: Calculating Proportions

To calculate the proportion of Democrats in a dataset, you would:

a) Count Democrats and divide by Republicans
b) Count Democrats and divide by total observations
c) Count total and divide by Democrats
d) Use mean() on a logical variable

Answer: **b) Count Democrats and divide by total observations** (Note: d is also correct if you have a logical variable indicating Democrats)

2.3.2 3.2 True or False: summarise()

Mark each statement as True (T) or False (F):

T summarise() reduces multiple rows to a single summary row **T** You can calculate multiple statistics in one summarise() call **F** summarise() automatically groups by all variables **T** n() counts the number of rows in each group **F** summarise() can only calculate numeric summaries

2.3.3 3.3 Fill in the Code

Complete this code to calculate turnout rate:

```
data %>%
  summarise(
    total_voters = n(),
    total_voted = sum(voted == "Yes"),
    turnout_rate = total_voted / total_voters
  )
```

2.4 4. Advanced Transformations

2.4.1 4.1 Match: Functions and Purposes

Match each function with its purpose:

Functions: a) round() b) log() c) sqrt() d) abs() e) lag()

Purposes: 1. Remove decimal places 2. Transform skewed data 3. Calculate square root 4. Make negative values positive 5. Get previous row's value

Matches: a = **1**, b = **2**, c = **3**, d = **4**, e = **5**

2.4.2 4.2 Code Detective: Complex Transformation

What does this code calculate?

```
data %>%
  group_by(state) %>%
  mutate(
    state_avg = mean(income, na.rm = TRUE),
    income_diff = income - state_avg,
    above_avg = income_diff > 0
  )
```

This code calculates: **For each state, it calculates the average income, then for each person it calculates how their income differs from their state's average, and creates a logical variable indicating if they earn above their state's average**

3 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: "Using tidyverse in R, create a new variable called ideology_group based on ideology_score. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]"

3.1 5. Creating Political Variables

Dataset: data/legislative_votes.csv

3.1.1 5.1 Initial Data Exploration

```r
# Load the dataset
legislators <- read_csv("data/legislative_votes.csv")
```

```
Rows: 500 Columns: 10
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): name, party, state
dbl (7): legislator_id, district, ideology_score, vote_attendance, bills_spo...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Examine the data
glimpse(legislators)
```

```
Rows: 500
Columns: 10
$ legislator_id   <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,~
$ name            <chr> "Legislator 1", "Legislator 2", "Legislator 3", "Legis~
```

```
$ party           <chr> "Democrat", "Democrat", "Republican", "Democrat", "Dem~
$ state           <chr> "MO", "AK", "PA", "FL", "HI", "CO", "NM", "CA", "IN", ~
$ district        <dbl> 17, 7, 10, 12, 4, 20, 3, 1, 16, 5, 19, 17, 11, 1, 1, 1~
$ ideology_score  <dbl> -0.765710472, 0.399555999, 0.117957905, 0.576843757, 0~
$ vote_attendance <dbl> 91.73352, 86.48068, 77.40871, 76.54805, 89.24765, 97.6~
$ bills_sponsored <dbl> 4, 8, 10, 5, 4, 2, 6, 8, 7, 5, 3, 7, 8, 10, 2, 7, 3, 4~
$ years_service   <dbl> 2, 36, 19, 36, 36, 22, 14, 11, 5, 9, 1, 8, 4, 29, 37, ~
$ committee_count <dbl> 1, 4, 4, 4, 6, 4, 8, 8, 5, 5, 3, 5, 4, 1, 6, 3, 8, 8, ~
```

3.1.2 5.2 Creating Categorical Variables

```r
# Create ideology categories based on terciles
legislators <- legislators %>%
  mutate(
    # Calculate tercile cutpoints for ideology
    ideology_category = case_when(
      ideology_score < quantile(ideology_score, 1/3, na.rm = TRUE) ~ "left",
      ideology_score < quantile(ideology_score, 2/3, na.rm = TRUE) ~ "center",
      TRUE ~ "right"
    ),

    # Create effectiveness score
    # Assume effective if sponsored more than 4 bills AND serves on committees
    effectiveness = if_else(bills_sponsored > 4 & committee_count > 0,
                            "effective", "not effective"),

    # Create seniority category
    seniority = if_else(years_service > 20, "senior", "junior")
  )

# Check the results
legislators %>%
  count(ideology_category)
```

```
# A tibble: 3 x 2
  ideology_category     n
  <chr>             <int>
1 center              166
2 left                167
3 right               167
```

```r
legislators %>%
  count(effectiveness)
```

```
# A tibble: 2 x 2
  effectiveness     n
  <chr>         <int>
1 effective       275
2 not effective   225
```

```r
legislators %>%
  count(seniority)
```

```
# A tibble: 2 x 2
  seniority     n
  <chr>     <int>
1 junior      258
2 senior      242
```

3.1.3 5.3 Calculating Party Metrics

```r
# Calculate party-level statistics for the created variables
party_stats <- legislators %>%
  group_by(party) %>%
  summarise(
    # Ideology distribution
    pct_left = mean(ideology_category == "left", na.rm = TRUE) * 100,
    pct_center = mean(ideology_category == "center", na.rm = TRUE) * 100,
    pct_right = mean(ideology_category == "right", na.rm = TRUE) * 100,

    # Effectiveness
    pct_effective = mean(effectiveness == "effective", na.rm = TRUE) * 100,

    # Seniority
    pct_senior = mean(seniority == "senior", na.rm = TRUE) * 100,

    n = n()
  ) %>%
  mutate(across(where(is.numeric) & !n, ~round(., 1)))

party_stats
```

```
# A tibble: 3 x 7
  party       pct_left pct_center pct_right pct_effective pct_senior     n
  <chr>          <dbl>      <dbl>     <dbl>         <dbl>      <dbl> <int>
1 Democrat        32.5       30.3      37.2          60.2       46.8   231
2 Independent     55         35        10            55         45      20
3 Republican      32.5       35.7      31.7          50.2       50.2   249
```

3.2 6. Education and Political Participation

Dataset: data/civic_engagement.csv

3.2.1 6.1 Loading and Initial Transformation

```r
# Load the dataset
civic <- read_csv("data/civic_engagement.csv")
```

```
Rows: 1000 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): education, voted_2020
dbl (7): respondent_id, age, income, political_interest, volunteer_hours, do...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Check the structure
glimpse(civic)
```

```
Rows: 1,000
Columns: 9
$ respondent_id          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, ~
$ education              <chr> "Master's", "High School", "Some College", "Hig~
$ age                    <dbl> 42, 36, 62, 81, 77, 77, 21, 56, 71, 24, 24, 66,~
$ income                 <dbl> 54146.01, 25746.07, 29484.80, 11927.46, 33049.0~
$ voted_2020             <chr> "Yes", "Yes", "No", "Yes", "Yes", "No", "Yes", ~
$ political_interest     <dbl> 8, 9, 7, 9, 6, 7, 8, 2, 5, 4, 1, 7, 2, 3, 4, 6,~
$ volunteer_hours        <dbl> 3, 1, 2, 1, 1, 1, 1, 1, 2, 1, 1, 5, 2, 2, 2, 1,~
$ donations              <dbl> 2417.21218, 169.08434, 15.33118, 0.00000, 62.12~
$ social_media_political <dbl> 7.8416122, 1.8931274, 7.4243669, 4.5703181, 2.0~
```

3.2.2 6.2 Education and Engagement Index

```
# Calculate medians for thresholds
median_donations <- median(civic$donations, na.rm = TRUE)
median_volunteer <- median(civic$volunteer_hours, na.rm = TRUE)

# Create new variables
civic <- civic %>%
  mutate(
    # Recode education into binary
    education_binary = if_else(
      education %in% c("High School", "Some College", "Bachelor", "Graduate"),
      "High School +",
      "Less than High School"
    ),

    # Create activist variable
    # TRUE if donated > median AND volunteered > median AND voted in 2020
    activist = (donations > median_donations) &
               (volunteer_hours > median_volunteer) &
               (voted_2020 == "Yes")
  )

# Check the results
civic %>%
  count(education_binary)
```

```
# A tibble: 2 x 2
  education_binary          n
  <chr>                 <int>
1 High School +           555
2 Less than High School   445
```

```
civic %>%
  count(activist)
```

```
# A tibble: 2 x 2
  activist     n
  <lgl>    <int>
1 FALSE      949
2 TRUE        51
```

```r
# Analyze relationship between education and activism
civic %>%
  group_by(education_binary) %>%
  summarise(
    pct_activist = mean(activist, na.rm = TRUE) * 100,
    avg_donations = mean(donations, na.rm = TRUE),
    avg_volunteer = mean(volunteer_hours, na.rm = TRUE),
    n = n()
  )
```

```
# A tibble: 2 x 5
  education_binary      pct_activist avg_donations avg_volunteer     n
  <chr>                        <dbl>         <dbl>         <dbl> <int>
1 High School +                 3.60          81.9          1.92   555
2 Less than High School         6.97          83.5          2.02   445
```

3.3 7. Campaign Finance Transformations

Dataset: data/campaign_finance_2024.csv

3.3.1 7.1 Data Preparation

```r
# Load the dataset
finance <- read_csv("data/campaign_finance_2024.csv")
```

```
Rows: 500 Columns: 10
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (7): candidate, party, office, contributor_type, state, employer, occup...
dbl  (2): contribution_id, amount
date (1): date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Explore the data
glimpse(finance)
```

```
Rows: 500
Columns: 10
$ contribution_id  <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16~
$ candidate        <chr> "Candidate I", "Candidate F", "Candidate E", "Candida~
$ party            <chr> "Democrat", "Republican", "Republican", "Democrat", "~
$ office           <chr> "Senate", "Governor", "Senate", "Governor", "Senate",~
$ amount           <dbl> 579.59202, 146.27368, 642.60921, 459.48787, 567.46356~
$ date             <date> 2024-04-18, 2024-01-12, 2024-03-06, 2024-08-30, 2024~
$ contributor_type <chr> "Individual", "Individual", "Individual", "Individual~
$ state            <chr> "WV", "VT", "HI", "AR", "OK", "GA", "KY", "IL", "RI",~
$ employer         <chr> "Tech Company", "University", "Tech Company", "Univer~
$ occupation       <chr> "Executive", "Physician", "Executive", "Retired", "Re~
```

3.3.2 7.2 Creating Contribution Categories

```r
# Calculate mean for threshold
mean_amount <- mean(finance$amount, na.rm = TRUE)

# Create donor categories
finance <- finance %>%
  mutate(
    # Donor size categories based on mean
    donor_size = if_else(
      amount < mean_amount,
      "small_donor",
      "major_donor"
    ),

    # In-state vs out-of-state (assuming we're in New York)
    contribution_source = if_else(
      state == "NY",
      "in_state",
      "out_of_state"
    )
  )

# Summarize results
finance %>%
  count(donor_size) %>%
  mutate(pct = n / sum(n) * 100)
```

```
# A tibble: 2 x 3
  donor_size      n   pct
  <chr>       <int> <dbl>
1 major_donor   128  25.6
2 small_donor   372  74.4
```

```
finance %>%
  count(contribution_source) %>%
  mutate(pct = n / sum(n) * 100)
```

```
# A tibble: 2 x 3
  contribution_source     n   pct
  <chr>               <int> <dbl>
1 in_state                9   1.8
2 out_of_state          491  98.2
```

```
# Analyze by party
finance %>%
  group_by(party, donor_size) %>%
  summarise(
    total_raised = sum(amount, na.rm = TRUE),
    avg_contribution = mean(amount, na.rm = TRUE),
    n_contributions = n(),
    .groups = "drop"
  )
```

```
# A tibble: 4 x 5
  party      donor_size  total_raised avg_contribution n_contributions
  <chr>      <chr>              <dbl>            <dbl>           <int>
1 Democrat   major_donor      192644.            3107.              62
2 Democrat   small_donor       63420.             341.             186
3 Republican major_donor      203661.            3086.              66
4 Republican small_donor       59756.             321.             186
```

3.3.3 7.3 Time-Based Transformations

```
# Convert date and create time variables
finance <- finance %>%
  mutate(
```

```r
    date = as.Date(date),
    month = month(date, label = TRUE),
    month_num = month(date),

    # Create campaign period categories
    campaign_period = case_when(
      month_num <= 3 ~ "early",
      month_num <= 7 ~ "middle",
      TRUE ~ "late"
    )
  )

# When was the most money given?
finance %>%
  group_by(campaign_period) %>%
  summarise(
    total = sum(amount, na.rm = TRUE),
    avg = mean(amount, na.rm = TRUE),
    n = n()
  ) %>%
  arrange(desc(total))
```

```
# A tibble: 3 x 4
  campaign_period   total   avg     n
  <chr>             <dbl> <dbl> <int>
1 middle          245911. 1128.   218
2 early           150857.  999.   151
3 late            122713.  937.   131
```

```r
# Monthly totals
monthly_totals <- finance %>%
  group_by(month) %>%
  summarise(
    total_amount = sum(amount, na.rm = TRUE),
    n_contributions = n(),
    avg_contribution = mean(amount, na.rm = TRUE)
  )

monthly_totals
```

```
# A tibble: 10 x 4
```

```
   month total_amount n_contributions avg_contribution
   <ord>         <dbl>           <int>            <dbl>
 1 Jan          58764.              53            1109.
 2 Feb          56757.              53            1071.
 3 Mar          35336.              45             785.
 4 Apr          43241.              60             721.
 5 May          58630.              49            1197.
 6 Jun          48439.              50             969.
 7 Jul          95601.              59            1620.
 8 Aug          35444.              46             771.
 9 Sep          46788.              48             975.
10 Oct          40481.              37            1094.
```

```r
# Cumulative contributions over time
finance %>%
  arrange(date) %>%
  mutate(cumulative_total = cumsum(amount)) %>%
  group_by(month) %>%
  summarise(
    cumulative = max(cumulative_total),
    month_total = sum(amount)
  )
```

```
# A tibble: 10 x 3
   month cumulative month_total
   <ord>      <dbl>       <dbl>
 1 Jan       58764.      58764.
 2 Feb      115521.      56757.
 3 Mar      150857.      35336.
 4 Apr      194098.      43241.
 5 May      252728.      58630.
 6 Jun      301167.      48439.
 7 Jul      396767.      95601.
 8 Aug      432212.      35444.
 9 Sep      479000.      46788.
10 Oct      519481.      40481.
```

3.4 8. Demographic Transformations

Dataset: data/census_political.csv

3.4.1 8.1 Initial Exploration

```r
# Load the dataset
census <- read_csv("data/census_political.csv")
```

```
Rows: 500 Columns: 10
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): county_name, state
dbl (8): county_id, population, median_age, pct_college, median_income, unem...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Look at the data
glimpse(census)
```

```
Rows: 500
Columns: 10
$ county_id          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,~
$ county_name        <chr> "County 1", "County 2", "County 3", "County 4", "C~
$ state              <chr> "NE", "KY", "KY", "CT", "RI", "KY", "WV", "IA", "A~
$ population         <dbl> 283363, 365881, 612761, 700633, 228297, 982323, 14~
$ median_age         <dbl> 42.17757, 39.16953, 47.61192, 32.49899, 33.38091, ~
$ pct_college        <dbl> 43.18251, 42.55179, 16.03140, 35.61717, 39.63398, ~
$ median_income      <dbl> 80961.13, 54125.69, 56570.75, 68557.12, 38715.02, ~
$ unemployment_rate  <dbl> 7.517104, 4.788997, 8.653567, 4.613327, 2.577154, ~
$ dem_vote_share_2020 <dbl> 46.79610, 42.95137, 48.17567, 25.27829, 46.37056, ~
$ turnout_2020       <dbl> 54.75303, 75.73557, 58.97276, 75.85745, 77.93041, ~
```

3.4.2 8.2 Creating Composite Indicators

```r
# Calculate median population for urban/rural classification
median_pop <- median(census$population, na.rm = TRUE)

# Create classifications
census <- census %>%
  mutate(
```

```r
    # Urban/rural classification based on population (proxy for density)
    urban_rural = if_else(
      population > median_pop,
      "urban",
      "rural"
    ),

    # Unemployment quartiles
    unemployment_quartile = case_when(
      unemployment_rate <= quantile(unemployment_rate, 0.25, na.rm = TRUE) ~ "Q1 (lowest)",
      unemployment_rate <= quantile(unemployment_rate, 0.50, na.rm = TRUE) ~ "Q2",
      unemployment_rate <= quantile(unemployment_rate, 0.75, na.rm = TRUE) ~ "Q3",
      TRUE ~ "Q4 (highest)"
    )
  )

# Check distributions
census %>%
  count(urban_rural) %>%
  mutate(pct = n / sum(n) * 100)
```

```
# A tibble: 2 x 3
  urban_rural     n   pct
  <chr>       <int> <dbl>
1 rural         250    50
2 urban         250    50
```

```r
census %>%
  count(unemployment_quartile)
```

```
# A tibble: 4 x 2
  unemployment_quartile     n
  <chr>                 <int>
1 Q1 (lowest)             125
2 Q2                      125
3 Q3                      125
4 Q4 (highest)            125
```

```r
# Analyze political patterns by urban/rural
census %>%
```

```r
  group_by(urban_rural) %>%
  summarise(
    avg_dem_share = mean(dem_vote_share_2020, na.rm = TRUE),
    avg_turnout = mean(turnout_2020, na.rm = TRUE),
    avg_college = mean(pct_college, na.rm = TRUE),
    avg_income = mean(median_income, na.rm = TRUE),
    n = n()
  )
```

```
# A tibble: 2 x 6
  urban_rural avg_dem_share avg_turnout avg_college avg_income     n
  <chr>               <dbl>       <dbl>       <dbl>      <dbl> <int>
1 rural                48.2        65.3        35.7     58804.   250
2 urban                50.6        65.7        38.1     62370.   250
```

```r
# Analyze by unemployment quartiles
census %>%
  group_by(unemployment_quartile) %>%
  summarise(
    avg_dem_share = mean(dem_vote_share_2020, na.rm = TRUE),
    avg_turnout = mean(turnout_2020, na.rm = TRUE),
    avg_income = mean(median_income, na.rm = TRUE),
    avg_college = mean(pct_college, na.rm = TRUE),
    n = n()
  ) %>%
  arrange(unemployment_quartile)
```

```
# A tibble: 4 x 6
  unemployment_quartile avg_dem_share avg_turnout avg_income avg_college     n
  <chr>                         <dbl>       <dbl>      <dbl>       <dbl> <int>
1 Q1 (lowest)                    49.8        65.3     59067.        36.8   125
2 Q2                             48.1        65.6     61377.        37.0   125
3 Q3                             50.8        65.2     58443.        36.6   125
4 Q4 (highest)                   48.7        65.9     63462.        37.2   125
```

4 Segment B — Causality

5 Non-AI Exercises

5.1 1. Three Requirements for Causality

5.1.1 1.1 Match: Causal Requirements

Match each requirement with its definition:

Requirements: a) Association b) Temporal ordering c) No confounding

Definitions: 1. The cause must come before the effect 2. No third variable explains both X and Y 3. X and Y must be statistically related

Matches: a = 3, b = 1, c = 2

5.1.2 1.2 Multiple Choice: Ice Cream and Crime

Why is "ice cream sales cause crime" a spurious correlation?

a) Ice cream and crime are not associated
b) Crime happens before ice cream sales
c) Temperature affects both ice cream sales and crime
d) The correlation is too weak

Answer: c

Explanation: Temperature is a confounding variable that increases both ice cream sales (hot weather) and crime (more people outside, more opportunities).

5.1.3 1.3 True or False: Causation

Mark each statement as True (T) or False (F):

F Correlation always implies causation **T** Causation requires temporal ordering **F** Large correlations are more likely to be causal **T** Randomized experiments help establish causation **F** Observational data can never show causation

Notes: - Correlation is necessary but not sufficient for causation - Correlation strength doesn't indicate causality - even strong correlations can be spurious - With proper controls and natural experiments, observational data can provide causal evidence

5.1.4 1.4 Multiple Choice: What Correlation Alone Can Tell Us

We find that counties with more coffee shops have higher voter turnout. Even before establishing whether coffee shops *cause* turnout, this correlation is still useful because it lets us:

a) Conclude that opening more coffee shops would raise turnout
b) Predict turnout — knowing a county's coffee-shop density helps us anticipate its turnout
c) Recommend coffee shops as a get-out-the-vote strategy
d) Rule out any confounding variables

Answer: b

Explanation: Correlation answers "what goes together?" — it supports prediction and screening even without a causal mechanism. Options a and c assume causation ("what happens if we intervene?"), which correlation alone cannot establish; d is false, since correlation does nothing to rule out confounders.

5.2 2. Identifying Confounders

5.2.1 2.1 Fill in the Blanks: Confounding

A confounding variable:

1. Affects both the **treatment** and the **outcome**
2. Creates a **spurious** relationship between X and Y
3. Must occur **prior** to both X and Y
4. Can be controlled through **random** assignment
5. Makes causal inference **difficult**

Word bank: treatment, outcome, spurious, prior, random, difficult

5.2.2 Match: Examples

Match each scenario with the likely confounding variable:

Scenarios: a) Wealthy people live longer b) Students with more books at home do better in school c) Cities with more police have more crime d) Married people are happier

Confounders: 1. Crime rates drive police hiring 2. Access to healthcare 3. Parents' education and income 4. Selection into marriage

Matches: a = 2, b = 3, c = 1, d = 4

Explanations: - a: Wealth → healthcare access → longevity - b: Parental education/income → books at home AND student performance - c: Reverse causation - high crime causes cities to hire more police - d: Selection bias - happier/more stable people more likely to marry

5.3 3. Temporal Ordering

5.3.1 3.1 Multiple Choice: Time Order

Which scenario violates temporal ordering for causation?

a) Studying causes better grades
b) Rain causes wet streets
c) High crime causes more police hiring last year
d) Tall children have tall parents

Answer: c

Explanation: The claim says high crime NOW caused police hiring LAST YEAR - the effect would precede the cause, violating temporal ordering.

5.3.2 3.2 Spot the Error: Causal Claims

What's wrong with this causal claim?

"States with stricter gun laws have more gun violence, so gun laws cause violence."

Problem: Reverse causation - states likely adopt stricter gun laws in response to existing gun violence problems. The high violence comes before the laws, not after.

5.3.3 3.3 Spot the Error: Post Hoc Reasoning

A mayor argues: "We adopted a new policing policy in 2021, and crime fell every year afterward — the policy cut crime." The data show crime had been falling by about the same amount each year *before* 2021 as well.

What reasoning error is this, and why does the timing alone not prove the policy worked?

Problem: This is the *post hoc, ergo propter hoc* fallacy — assuming that because the drop followed the policy, the policy caused it. Crime was already on a steady downward trend that continued unchanged through 2021, so the pre-existing trend explains the decline. Temporal ordering (the policy came first) is necessary but not sufficient for causation.

5.4 4. Alternative Explanations

5.4.1 4.1 Multiple Choice: Spurious Correlations

The correlation between divorce rate and margarine consumption is likely:

a) Causal - margarine causes divorce
b) Causal - divorce causes margarine consumption
c) Spurious - both follow similar time trends
d) Reversed - the causation goes backwards

Answer: c

Explanation: This is a classic spurious correlation. Both variables happened to decline over the same time period for unrelated reasons. No causal mechanism connects margarine to divorce.

5.4.2 4.2 True or False: Research Design

Mark each statement as True (T) or False (F):

T Random assignment eliminates all confounders **F** Large samples guarantee causal inference **T** Natural experiments can establish causation **T** Correlation is necessary but not sufficient for causation **F** Time series data always shows causation

Notes: - Random assignment eliminates both observed and unobserved confounders on average - Large samples improve precision but don't fix confounding or reverse causation - Natural experiments (e.g., draft lottery) can approximate random assignment - Time series can show correlations but temporal ordering doesn't prove causation

5.4.3 4.3 Match: Causal Mechanisms

Match each political example with its mechanism:

Examples: a) Education → Higher income b) Campaign spending → Votes c) Media coverage → Public opinion d) Voting laws → Turnout

Mechanisms: 1. Increased name recognition 2. Skills and credentials 3. Agenda setting and framing 4. Access and convenience

Matches: a = 2, b = 1, c = 3, d = 4

5.4.4 4.4 Multiple Choice: Selecting on the Outcome

A researcher studies **only** completed suicide-terror campaigns and finds that nearly all occurred under a foreign military occupation, concluding that occupation causes suicide terrorism. Why is this reasoning flawed?

a) Foreign occupation and suicide terrorism are not associated
b) By studying only cases where terrorism occurred (selecting on the outcome), the researcher never sees the many occupations that produced *no* terrorism
c) The correlation is too weak to interpret
d) Suicide terrorism causes occupation, not the other way around

Answer: b

Explanation: This is selection bias — selecting on the dependent variable. Studying only the attacks guarantees you "find" occupation everywhere. To test the claim you must also include occupations without terrorism; most produced none, so occupation is common but rarely produces suicide terrorism. (This is the standard critique of Robert Pape's occupation-and-terrorism study.)

6 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: "Using tidyverse in R, compare the average outcome between treatment and control groups. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]"

6.1 5. Testing Causal Claims

Dataset: data/minimum_wage_effects.csv

Description: State-level data on minimum wage changes and employment.

Variables:

- `state`: State name (chr)

- min_wage: State minimum wage (dbl)
- unemployment_rate: Unemployment percentage (dbl)
- teen_employment_rate: Teen employment percentage (dbl)
- cost_of_living: Cost of living index (dbl)
- gdp_growth: State GDP growth rate (dbl)
- border_state_wage: Average minimum wage in bordering states (dbl)

This is **cross-sectional** data: one row per state, no time dimension.

6.1.1 5.1 Initial Exploration

```
# Load the dataset
wage_data <- read_csv("data/minimum_wage_effects.csv")
```

```
Rows: 50 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): state
dbl (6): min_wage, unemployment_rate, teen_employment_rate, cost_of_living, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Examine the structure
glimpse(wage_data)
```

```
Rows: 50
Columns: 7
$ state                <chr> "Alabama", "Alaska", "Arizona", "Arkansas", "Cali~
$ min_wage             <dbl> 13.011488, 9.442253, 11.183902, 11.719136, 7.4994~
$ unemployment_rate    <dbl> 3.059832, 3.356263, 4.501135, 6.378855, 7.412944,~
$ teen_employment_rate <dbl> 30.10537, 47.93235, 45.38204, 53.89031, 30.32068,~
$ cost_of_living       <dbl> 95.40286, 98.16813, 99.23419, 103.79086, 98.32271~
$ gdp_growth           <dbl> 2.3824652, 6.4797684, 0.2158908, 2.8664880, 1.083~
$ border_state_wage    <dbl> 10.063658, 12.917990, 11.235054, 7.551498, 12.530~
```

```
# Summary statistics
summary(wage_data)
```

```
    state              min_wage      unemployment_rate teen_employment_rate
 Length:50          Min.   : 7.251   Min.   :3.060     Min.   :30.11
 Class :character   1st Qu.: 8.874   1st Qu.:4.638     1st Qu.:34.62
 Mode  :character   Median :10.232   Median :5.392     Median :40.47
                    Mean   :10.401   Mean   :5.602     Mean   :41.63
                    3rd Qu.:11.478   3rd Qu.:6.737     3rd Qu.:48.93
                    Max.   :14.991   Max.   :7.967     Max.   :54.28
 cost_of_living     gdp_growth       border_state_wage
 Min.   : 80.04   Min.   :-0.1086   Min.   : 7.307
 1st Qu.: 96.86   1st Qu.: 1.5158   1st Qu.: 9.653
 Median :105.50   Median : 2.3190   Median :10.829
 Mean   :110.32   Mean   : 2.6490   Mean   :10.891
 3rd Qu.:122.63   3rd Qu.: 3.9589   3rd Qu.:12.604
 Max.   :139.71   Max.   : 6.4798   Max.   :14.933
```

```r
# Check variation in minimum wage by whether the state has a
# high or low minimum wage relative to the national median
wage_data <- wage_data %>%
  mutate(wage_group = ifelse(min_wage >= median(min_wage),
                             "High wage", "Low wage"))

wage_data %>%
  group_by(wage_group) %>%
  summarize(
    min_wage_mean = mean(min_wage),
    min_wage_sd = sd(min_wage),
    n_states = n()
  )
```

```
# A tibble: 2 x 4
  wage_group min_wage_mean min_wage_sd n_states
  <chr>              <dbl>       <dbl>    <int>
1 High wage          12.0        1.39        25
2 Low wage            8.79       0.862       25
```

6.1.2 5.2 Checking Causal Requirements

Test for association between minimum wage and employment

```r
# 1. Association test
cor.test(wage_data$min_wage, wage_data$unemployment_rate)
```

```
	Pearson's product-moment correlation

data:  wage_data$min_wage and wage_data$unemployment_rate
t = 0.95981, df = 48, p-value = 0.342
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.1467263  0.4002841
sample estimates:
      cor 
0.1372258 
```

```
cor.test(wage_data$min_wage, wage_data$teen_employment_rate)
```

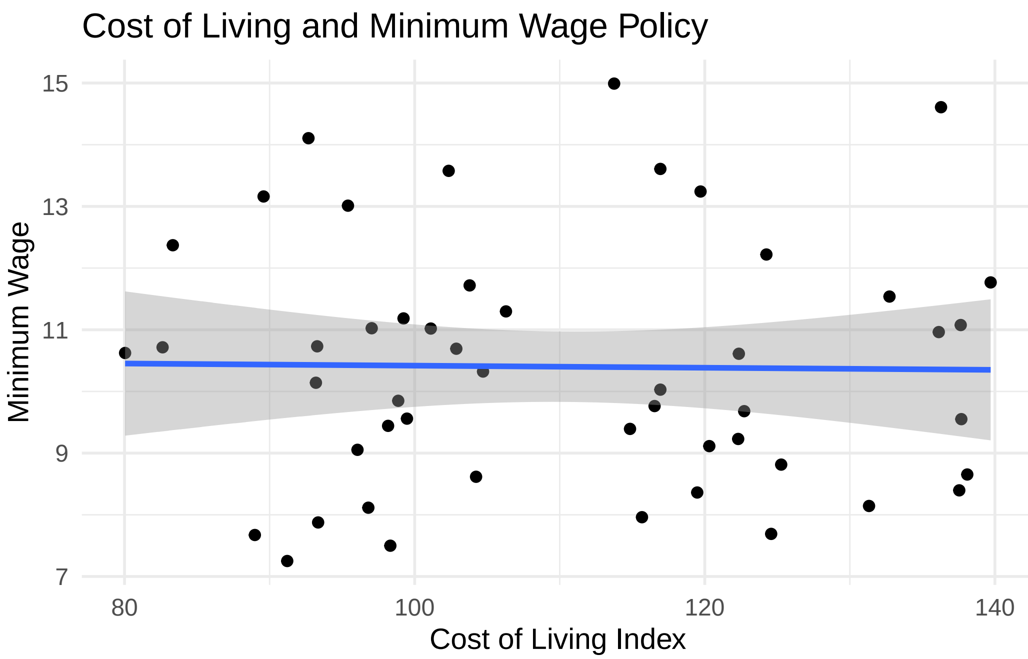
```
	Pearson's product-moment correlation

data:  wage_data$min_wage and wage_data$teen_employment_rate
t = 0.29188, df = 48, p-value = 0.7716
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.2390565  0.3167289
sample estimates:
       cor 
0.04209203 
```

Key Findings: - There IS an association between minimum wage and teen employment (negative)

6.1.3 5.3 Alternative Explanations

```
# Check if higher-cost states have higher minimum wages
# (cost of living is a plausible confounder)
ggplot(wage_data, aes(x = cost_of_living, y = min_wage)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "Cost of Living and Minimum Wage Policy",
       x = "Cost of Living Index",
       y = "Minimum Wage") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```r
# Is the state minimum wage related to wages in bordering states?
cor.test(wage_data$min_wage, wage_data$border_state_wage)
```

```

	Pearson's product-moment correlation

data:  wage_data$min_wage and wage_data$border_state_wage
t = -0.16425, df = 48, p-value = 0.8702
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.3000694  0.2563375
sample estimates:
        cor 
-0.02370132 
```

```r
cor.test(wage_data$min_wage, wage_data$cost_of_living)
```

```

	Pearson's product-moment correlation
```

```
data:  wage_data$min_wage and wage_data$cost_of_living
t = -0.10197, df = 48, p-value = 0.9192
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.2918690  0.2647152
sample estimates:
        cor
-0.01471686
```

Alternative Explanations: 1. **State characteristics**: Progressive states have higher wages AND different labor markets 2. **Economic conditions**: States with stronger growth (`gdp_growth`) may both raise wages and have different employment 3. **Cost of living**: Higher wage states also tend to have higher costs of living 4. **Regional spillovers**: Wages in neighboring states (`border_state_wage`) move together, so a state's wage is not set in isolation

6.2 6. Media Effects on Politics

Dataset: data/media_influence_study.csv

Description: Data on media consumption and political attitudes.

Variables:

- `respondent_id`: Unique identifier (int)
- `survey_wave`: Wave 1 or 2 (int)
- `fox_news_hours`: Weekly hours watching Fox News (dbl)
- `msnbc_hours`: Weekly hours watching MSNBC (dbl)
- `conservative_score`: Conservative attitudes, 0-100 (dbl)
- `trump_approval`: Trump approval, 0-100 (dbl)
- `political_interest`: Interest in politics, 1-10 (int)
- `party_id`: Party identification (chr)

6.2.1 6.1 Media and Attitudes

```r
# Load the dataset
media_data <- read_csv("data/media_influence_study.csv")
```

```
Rows: 500 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
```

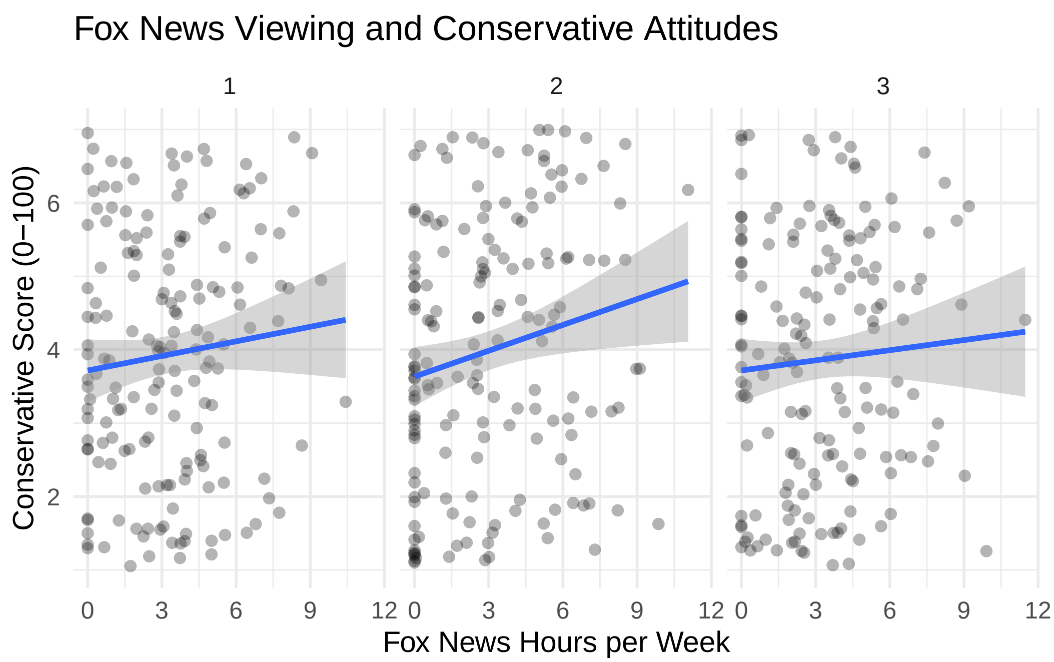
```
chr (1): party_id
dbl (7): respondent_id, survey_wave, fox_news_hours, msnbc_hours, conservati...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Check the structure
glimpse(media_data)
```

```
Rows: 500
Columns: 8
$ respondent_id      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ~
$ survey_wave        <dbl> 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, ~
$ fox_news_hours     <dbl> 1.7310839, 2.5611155, 1.0708562, 7.7472231, 0.00000~
$ msnbc_hours        <dbl> 5.9448073, 0.5349448, 6.8126954, 1.8358256, 3.87525~
$ conservative_score <dbl> 1.053381, 6.225898, 2.864188, 1.779599, 3.605809, 6~
$ trump_approval     <dbl> 50.57083, 48.65184, 72.31505, 27.02195, 42.74481, 5~
$ political_interest <dbl> 3, 6, 5, 3, 9, 6, 9, 7, 5, 3, 10, 5, 5, 8, 9, 10, 5~
$ party_id           <chr> "Independent", "Republican", "Independent", "Republ~
```

```r
# Examine relationship between Fox News and conservatism
ggplot(media_data, aes(x = fox_news_hours, y = conservative_score)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  facet_wrap(~survey_wave) +
  labs(title = "Fox News Viewing and Conservative Attitudes",
       x = "Fox News Hours per Week",
       y = "Conservative Score (0-100)") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



6.2.2 6.2 Reverse Causation

Test whether conservatives choose Fox News (selection)

```
# Compare Fox News viewing by party ID
media_data %>%
  filter(survey_wave == 1) %>%
  group_by(party_id) %>%
  summarize(
    mean_fox_hours = mean(fox_news_hours),
    mean_msnbc_hours = mean(msnbc_hours)
  )
```

```
# A tibble: 3 x 3
  party_id    mean_fox_hours mean_msnbc_hours
  <chr>                <dbl>            <dbl>
1 Democrat              3.00             2.30
2 Independent           3.19             2.36
3 Republican            3.48             2.30
```

Key Findings: - Republicans watch MORE Fox News than Democrats - This suggests **reverse causation**: conservatives choose Fox News (not just that Fox makes people conservative)

6.2.3 6.3 Confounding Factors

```
# Check if party ID affects both Fox viewing AND conservatism
media_data %>%
  filter(survey_wave == 1) %>%
  group_by(party_id) %>%
  summarize(
    mean_fox_hours = mean(fox_news_hours),
    mean_conservative_score = mean(conservative_score)
  )
```

```
# A tibble: 3 x 3
  party_id    mean_fox_hours mean_conservative_score
  <chr>                <dbl>                   <dbl>
1 Democrat              3.00                    3.80
2 Independent           3.19                    3.97
3 Republican            3.48                    4.01
```

Key Findings: - **Party ID is a confounder**: Republicans watch more Fox AND have higher conservative scores - This makes it hard to tell if Fox causes conservatism, or if party ID causes both - Party affiliation affects both the "treatment" (Fox viewing) and "outcome" (conservatism)

6.3 7. Campaign Finance and Elections

Dataset: data/campaign_causality.csv

Description: Congressional election data with spending information.

Variables:

- `district_id`: Congressional district (chr)
- `incumbent_party`: Party of incumbent (chr)
- `incumbent_spending`: Incumbent campaign spending (dbl)
- `challenger_spending`: Challenger campaign spending (dbl)
- `incumbent_vote_share`: Incumbent's vote percentage (dbl)
- `presidential_vote`: Presidential vote in district (dbl)

- `incumbent_scandal`: Whether incumbent had scandal (lgl)
- `quality_challenger`: Whether challenger held prior office (lgl)

6.3.1 7.1 Money and Votes

```
# Load the dataset
campaign_data <- read_csv("data/campaign_causality.csv")
```

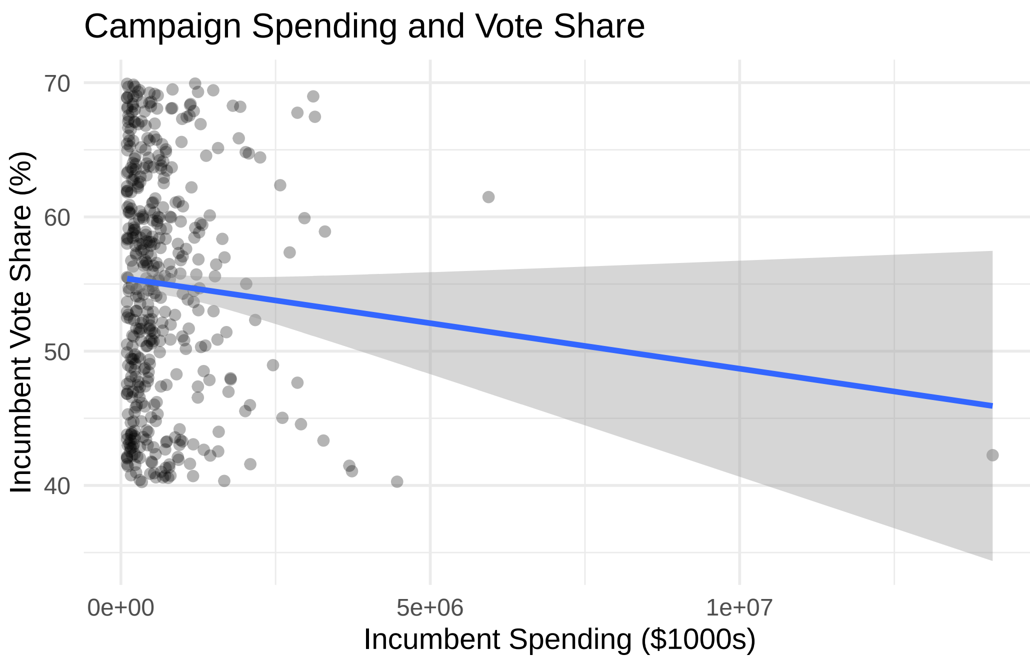
```
Rows: 435 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): incumbent_party
dbl (5): district_id, incumbent_spending, challenger_spending, incumbent_vot...
lgl (2): incumbent_scandal, quality_challenger

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Explore the data
glimpse(campaign_data)
```

```
Rows: 435
Columns: 8
$ district_id          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15~
$ incumbent_party      <chr> "Democrat", "Republican", "Republican", "Republic~
$ incumbent_spending   <dbl> 815245.9, 469650.7, 168369.3, 519741.0, 251934.8,~
$ challenger_spending  <dbl> 247902.86, 147862.05, 201944.42, 104452.86, 57408~
$ incumbent_vote_share <dbl> 55.90956, 60.55175, 48.11705, 52.92366, 52.98682,~
$ presidential_vote    <dbl> 36.01053, 33.32993, 69.36311, 53.49567, 59.91367,~
$ incumbent_scandal    <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, ~
$ quality_challenger   <lgl> FALSE, TRUE, FALSE, TRUE, FALSE, FALSE, FALSE, FA~
```

```
# Relationship between spending and vote share
ggplot(campaign_data, aes(x = incumbent_spending, y = incumbent_vote_share)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(title = "Campaign Spending and Vote Share",
       x = "Incumbent Spending ($1000s)",
       y = "Incumbent Vote Share (%)") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



6.3.2 7.2 Endogeneity Problems

Understanding endogeneity in campaign spending:

```
# 1. Do successful candidates raise more money?
# Check if expected competitiveness affects fundraising
campaign_data <- campaign_data %>%
  mutate(
    district_partisan_lean = presidential_vote - 50,
    competitive_race = abs(incumbent_vote_share - 50) < 10
  )

# Spending in competitive vs safe races
campaign_data %>%
  group_by(competitive_race) %>%
  summarize(
    avg_incumbent_spending = mean(incumbent_spending),
    avg_challenger_spending = mean(challenger_spending)
  )
```

```
# A tibble: 2 x 3
  competitive_race avg_incumbent_spending avg_challenger_spending
  <lgl>                             <dbl>                   <dbl>
1 FALSE                           643955.                 351148.
2 TRUE                            713829.                 280675.
```

```
# 2. Quality challengers raise more money
campaign_data %>%
  group_by(quality_challenger) %>%
  summarize(
    avg_challenger_spending = mean(challenger_spending),
    avg_incumbent_vote = mean(incumbent_vote_share)
  )
```

```
# A tibble: 2 x 3
  quality_challenger avg_challenger_spending avg_incumbent_vote
  <lgl>                                <dbl>              <dbl>
1 FALSE                              308071.               54.7
2 TRUE                               288131.               55.8
```

```
# 3. Control for confounders
controlled_model <- lm(incumbent_vote_share ~ incumbent_spending +
                       challenger_spending + presidential_vote +
                       incumbent_scandal + quality_challenger,
                       data = campaign_data)
summary(controlled_model)
```

```
Call:
lm(formula = incumbent_vote_share ~ incumbent_spending + challenger_spending +
    presidential_vote + incumbent_scandal + quality_challenger,
    data = campaign_data)

Residuals:
     Min       1Q   Median       3Q      Max
-16.2082  -7.2903   0.3306   7.0920  15.5138

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)             5.650e+01  1.947e+00  29.013   <2e-16 ***
incumbent_spending     -7.385e-07  4.401e-07  -1.678   0.0941 .
```

```
challenger_spending     2.862e-07  8.537e-07   0.335   0.7376
presidential_vote      -2.815e-02  3.597e-02  -0.783   0.4343
incumbent_scandalTRUE   9.854e-01  1.678e+00   0.587   0.5574
quality_challengerTRUE  1.073e+00  9.476e-01   1.132   0.2584
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.705 on 429 degrees of freedom
Multiple R-squared:  0.01133,   Adjusted R-squared:  -0.0001938
F-statistic: 0.9832 on 5 and 429 DF,  p-value: 0.4276
```

Endogeneity Problems:

1. **Expected competitiveness**: Close races attract more spending from both sides
2. **Reverse causation**: Popular candidates raise more money
3. **Quality challengers**: Strong challengers raise more money AND do better
4. **Omitted variables**: District partisanship, candidate quality

Effect of controls: - Spending effect gets SMALLER when controlling for confounders - Suggests much of the correlation is spurious

6.3.3 7.3 Natural Experiments

Natural Experiment Ideas: 1. **Scandals**: Create unexpected competitive races, forcing increased spending 2. **Term limits**: Open seats with no incumbent advantage 3. **Matching**: Compare similar districts with different spending levels 4. **Instrumental variables**: Use distance to major donors as instrument for spending

6.4 8. Policy Evaluation

Dataset: data/policy_intervention.csv

Description: Data on states before/after policy changes.

Variables:

- `state`: State name (chr)
- `policy_adopted`: Whether state adopted policy (lgl)
- `year_adopted`: Year the policy was (or would have been) adopted (int)
- `outcome_variable`: Policy outcome of interest (dbl)
- `population`: State population (int)
- `unemployment`: Unemployment rate (dbl)
- `gov_party`: Governor's party (chr)

- `leg_control`: Party controlling legislature (chr)
- `neighbor_adopted`: Whether neighboring state has policy (lgl)

This is **cross-sectional** data: one row per state.

6.4.1 8.1 Policy Effects

```r
# Load the dataset
policy_data <- read_csv("data/policy_intervention.csv")
```

```
Rows: 50 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): state, gov_party, leg_control
dbl (4): year_adopted, outcome_variable, population, unemployment
lgl (2): policy_adopted, neighbor_adopted

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Look at the structure
glimpse(policy_data)
```

```
Rows: 50
Columns: 9
$ state            <chr> "Alabama", "Alaska", "Arizona", "Arkansas", "Californ~
$ policy_adopted   <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, TRUE, FALSE, TRUE,~
$ year_adopted     <dbl> 2019, 2010, 2013, 2022, 2010, 2011, 2012, 2020, 2012,~
$ outcome_variable <dbl> 44.38602, 28.55397, 69.12784, 52.56511, 35.59081, 40.~
$ population       <dbl> 22497078, 27981889, 37781574, 1856289, 31414300, 9333~
$ unemployment     <dbl> 8.999175, 4.260362, 6.680870, 7.767551, 3.196815, 8.2~
$ gov_party        <chr> "Republican", "Republican", "Republican", "Republican~
$ leg_control      <chr> "Split", "Republican", "Republican", "Democrat", "Rep~
$ neighbor_adopted <lgl> TRUE, TRUE, FALSE, FALSE, FALSE, TRUE, TRUE, TRUE, FA~
```

```r
# Compare outcomes: states with vs without policy
policy_data %>%
  group_by(policy_adopted) %>%
  summarize(
```

```r
    mean_outcome = mean(outcome_variable),
    n = n()
  )
```

```
# A tibble: 2 x 3
  policy_adopted mean_outcome     n
  <lgl>                 <dbl> <int>
1 FALSE                  50.7    32
2 TRUE                   49.7    18
```

Key Findings: - States WITH the policy have higher outcomes than states WITHOUT - But is this causal? Or do certain types of states adopt the policy?

6.4.2 8.2 Selection into Treatment

Check whether adopting states differ on background characteristics that pre-date the policy

```r
# Compare adopters vs non-adopters on a likely pre-existing
# characteristic (unemployment) that the policy did not cause
policy_data %>%
  group_by(policy_adopted) %>%
  summarize(
    mean_outcome = mean(outcome_variable),
    mean_unemployment = mean(unemployment),
    n = n()
  )
```

```
# A tibble: 2 x 4
  policy_adopted mean_outcome mean_unemployment     n
  <lgl>                 <dbl>             <dbl> <int>
1 FALSE                  50.7              6.47    32
2 TRUE                   49.7              7.13    18
```

Key Findings: - States that adopt the policy differ on background characteristics (e.g., unemployment) - This is **selection bias** - certain types of states choose to adopt - Because these differences are not caused by the policy, they make it hard to tell whether the policy caused the outcome gap or whether the gap was pre-existing

6.4.3 8.3 Confounders in Policy Adoption

Check if political factors predict which states adopt

```r
# Compare political characteristics of adopters vs non-adopters
policy_data %>%
  group_by(policy_adopted) %>%
  summarize(
    pct_dem_governor = mean(gov_party == "Democrat"),
    pct_dem_legislature = mean(leg_control == "Democrat")
  )
```

```
# A tibble: 2 x 3
  policy_adopted pct_dem_governor pct_dem_legislature
  <lgl>                     <dbl>               <dbl>
1 FALSE                     0.375               0.281
2 TRUE                      0.389               0.333
```

Key Findings: - States with Democratic governors and legislatures are MORE likely to adopt the policy - **Political ideology is a confounder**: it affects BOTH policy adoption AND outcomes - Makes causal inference difficult - is it the policy or the political context?

Main Lesson: - Simply comparing states with and without the policy gives misleading results - Need to account for selection bias, confounders, and pre-existing differences - Advanced methods (difference-in-differences, matching) can help but require careful assumptions